Gregor Bučar

Senior project engineer in software and automation for GMP pharmaceutical manufacturing

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SUMMARY

I build software in C# and .NET, from desktop apps and web front ends to the services and APIs behind them. Most of what I've built runs in pharmaceutical plants, where the rules are strict and everything has to be validated and traceable. The recipe editor I wrote most of is used at hundreds of installations. My software usually talks to real machines, so I also write PLC code when a project needs it, along with the specifications and test protocols a regulated plant requires.

SKILLS AT A GLANCE

C# and .NET	•••• Expert, since 2018	WPF, WinUI 3, ASP.NET Core, EF Core, SignalR
Desktop apps with MVVM	•••• Expert, since 2020	a process database application with a recipe editor, used at hundreds of plants
TypeScript and Node.js	••••○ Advanced, since 2022	Fastify services between the PLC and the browser
Vue and Nuxt	••••○ Advanced, since 2023	operator apps and planning tools, with Tailwind
Automated testing	••••○ Advanced, since 2022	xUnit, Vitest, hardware-in-the-loop against a real PLC
Deployment and hosting	••••○ Advanced, since 2024	Docker, GitHub Actions, Cloudflare, Linux and Windows services
Computer vision	••••○ Advanced, since 2023	training YOLO models, OpenCV, Python
Building with coding agents	••••○ Advanced, since 2025	Claude Code, with a spec and a plan first, tests and my review
PLC programming	••••○ Advanced, since 2024	object-oriented Structured Text in CODESYS, SCL in TIA Portal
ISA-88 batch control	••••○ Advanced, since 2020	PLC phases and the recipe software above them
OPC UA and fieldbuses	••••○ Advanced, since 2022	PROFINET, Modbus RTU and TCP, RS-232, RS-485, EtherNet/IP
GxP validation	••••○ Advanced, since 2022	URS, FS, SDS, OQ, installation qualification, audit trails
Home automation	••••○ Advanced, 2026	Home Assistant APIs, a custom integration, voice, WLED

Expert means I have built and shipped production systems with it for years, **Advanced** means I have shipped production work with it on my own, **Working** means I have used it in real projects, and **Basics** means I have set it up or used it a few times.

EXPERIENCE

Senior Project Engineer, *Trac d.o.o., Šentjernej* — Jun 2024 – present

I worked at Trac as a student from 2018 to May 2024. Trac builds automation, batch-management software and lab robots for pharmaceutical plants.

- **Batch-manufacturing platform rewrite** (.NET 10, 2026, with coding agents). I'm the top contributor. I rebuilt the recipe engine with a state machine per step, crash recovery and a replay queue, and wrote hardware-in-the-loop tests against a real PLC, traced to the functional specification and OQ. I also wrote the approved URS, FS, SDS, HDS and OQ documents for the engine.
- **Recipe editor** (WPF, 2020 to 2025). I wrote most of the sequential-function-chart editor that process engineers use to draw batch recipes, including the condition builder. It runs at hundreds of installations with thousands of users.

- **Lab sample-preparation robot** (2023 to 2026). I wrote nearly all the module software (six Node.js/TypeScript services on one shared core, Nuxt operator screens in four languages, the audit trail, fleet deploys and central logging) and most of the PLC module programs on seven CODESYS controllers, with PROFINET servos, EtherNet/IP step motors, a Modbus RTU gripper and my own RS-232/RS-485 serial library. I also built the YOLO model that finds the sample tubes and the game-controller arm control, and moved the controllers to a real-time Linux kernel, which cut PROFINET worst-case latency from 301 to 94 µs.
- **Installation-qualification tool** (WinUI 3, ASP.NET Core, OPC UA, 2022 to 2026). I wrote about 90% of the desktop app and three quarters of the API. It tests every PLC signal live, records who tested it and the result, and has been used on five large projects.
- **Siemens and building automation** (2026). I wrote seven of the eight ISA-88 cleaning operations for an IBC washer on an S7-1500 in SCL, with my own code generators and checker, and added climate sensors and device health checks to an office building's existing CODESYS program.
- **Workload planner** (Vue, .NET). It plans every task against people's real capacity six months ahead. It replaced a paid planning tool for 29 people and saves about €5,200 a year.
- I train customers' operators on site and work with their IT on server setup. I mentor two students, and from October 2026 I'm also on the support team.

Full-stack developer, student work, *Špica International* Sep 2022 – May 2023

- On a time-tracking SaaS I moved whole screens from AngularJS to Angular with NgRx and built ASP.NET Core endpoints with xUnit tests for the onboarding, pricing and trial screens.

EDUCATION

BSc in Computer and Information Science, Faculty of Computer and Information Science, University of Ljubljana 2019 – 2023

My thesis, *Design and implementation of a decentralised digital building logbook* (2023, mentor prof. dr. Vlado Stankovski), was graded 10 out of 10.

Technical gymnasium (electrical engineering and computing), Šolski center Novo mesto 2015 – 2019

3rd prize and silver award, national ACM computing competition for secondary schools (RTK), 2018, where I placed 6th of 99 in the first group.

OWN PROJECTS

- **Time-tracking and invoicing SaaS** (2026, with coding agents). It has a multi-tenant .NET 10 API with clean architecture, a Nuxt web app, an offline-first Expo mobile app, a tray timer and a browser extension. Tests fail the build if customers' data could mix, and it's deployed with Docker, GitHub Actions and Cloudflare. Free to use.
- **WebSCADA, now EVA** (2024 to 2026, team of five). I wrote most of the gateway to the field equipment (protocol plugins, keeps watching when the server is down), the licensing, a rule engine and the floor-plan editor. It runs at several sites.
- **Home automation** (2026). I run Home Assistant on a Raspberry Pi with my own voice assistant, a custom integration, WLED and local Tuya devices.

LANGUAGES AND MORE

Slovenian (native), English (C2), French (A2 to B1). Driving licence B.

The full version of this CV, with every project in detail, is at realgregor.github.io.